YUNUS EMRE ULUÇ

Software Engineer · AI & Python Backend

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SUMMARY

Software Engineering graduate (2026), **ranked first in both my faculty and department**. I build AI applications and backend systems in Python: a TÜBİTAK-funded diagnostic model trained on ~67,000 X-rays (**91.54% accuracy**) with a clinician web app in **FractureAI**, and a 36-tool LLM agent with a tested MCP server in **Chiron**. Looking for a **junior AI / Python backend** role where models ship as working products, with APIs, interfaces and tests.

PROJECTS

Chiron — LLM agent platform & MCP server GitHub 2026

- **36-tool function-calling agent** on the Gemini Live API: tool schemas, routing, error handling and multi-step tasks.
- **MCP server** on the official Python SDK, covered by **42 pytest tests** through a real MCP client; connected to Claude Code.
- FastAPI + WebSocket backend, Next.js UI, on-device YOLO-World and OCR, two-step approval for irreversible actions; Kotlin Android client.

FractureAI — TÜBİTAK 2209-A research project · project lead GitHub 2025 – 2026

- **~67,000 X-rays** from 4 open datasets; patient-level splits for a leakage-free training pipeline.
- EfficientNet-B5, ConvNeXt and Swin V2 ensemble: **91.54% accuracy, Cohen's κ 0.831, AUC 0.960**.
- FastAPI clinician web app with Grad-CAM explanations; project report and thesis.

Dilcep — offline translation & language-learning app (Android) 2026

- Expo / React Native with **3 native Kotlin modules** (ML Kit): text, speech and camera translation without internet; 191 lessons, SQLite + Supabase sync.

Dengue Fever Detection — machine learning study · under review GitHub 2026

- Compared 9 algorithms (XGBoost, Random Forest, SVM, MLP…) on 1,523 patient records with Stratified K-Fold.

EXPERIENCE

Software Engineering Intern — Opak Yazılım, Konya Jul 2025

- Built **NICEO** (Flutter) and **NEO** (React Native) mobile e-commerce apps: catalogue, cart, checkout, order tracking.
- Node.js / Express REST API (JWT), MySQL schema, natural-language product search with a local LLM (Ollama), Next.js admin panel.

EDUCATION

B.Sc. Software Engineering — Muş Alparslan University 2022 – 2026

Ranked first in faculty and department. Thesis: deep learning system for fracture and dislocation detection in clinical X-rays.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Kotlin, SQL
AI / ML	PyTorch, scikit-learn, LLM agents, function calling, MCP, Ollama, OpenCV, ONNX Runtime
Backend & Web	FastAPI, Node.js / Express, REST, WebSocket, PostgreSQL, MySQL, Next.js, React
Mobile	Kotlin / Jetpack Compose, React Native / Expo, Flutter
Tools	Git, pytest, Docker, Claude Code, Codex CLI

CERTIFICATION & LANGUAGES

CCNA: Introduction to Networks (Cisco, 2025) · Turkish: native · English: B1